Decision Making Under Uncertainty

MGMT 670: Business Analytics

PURDUE UNIVERSITY | Mitchell E. Daniels, Jr. School of Business

Risk

All interesting choices boil down to:

A sure thing vs. A gamble

- Sure Thing: pays $5
- Gamble:
  - pays $10 0.5 chance How about 0.4
  - or pays $0 0.5 chance 0.6

Examples

Sure Thing:
- 200K in certificate of deposit (CD) for 5 years at 2% per year

Gamble:
- Buy a house for 200K sell it in 5 years.

Sure Thing:
- Stay in current job.

Gamble:
- Go to Graduate School

Basic elements

- CHOICES versus CHANCES
  choice: stay at current job/ go to grad school
  chance: getting a high-paid job after master
- CONSEQUENCES (OUTCOMES)
- LIKELIHOOD(PROBABILITY)

This class: Decision Analysis

Goal: Find optimal strategies when there are

– several decision alternatives

– uncertain and risks associated with future events.

Using sample information (market test) to improve decision

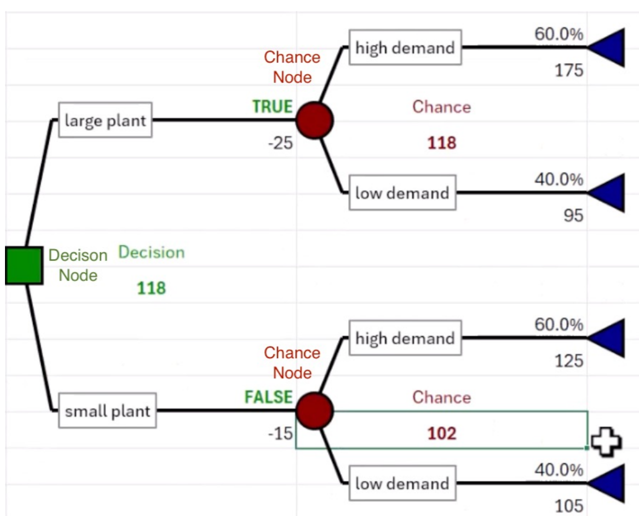
Important tool: Decision tree

Part 1: Decision at Colonial Motors

- Colonial Motors (CM) needs to determine whether to build a large or small plant for a new car it is developing. The cost of constructing a large plant is $25 million and the cost of constructing a small plant is $15 million.
- CM believes there is a 60% chance that demand for the new car will be high and a 40% chance that it will be low. Here is the sales (in million):

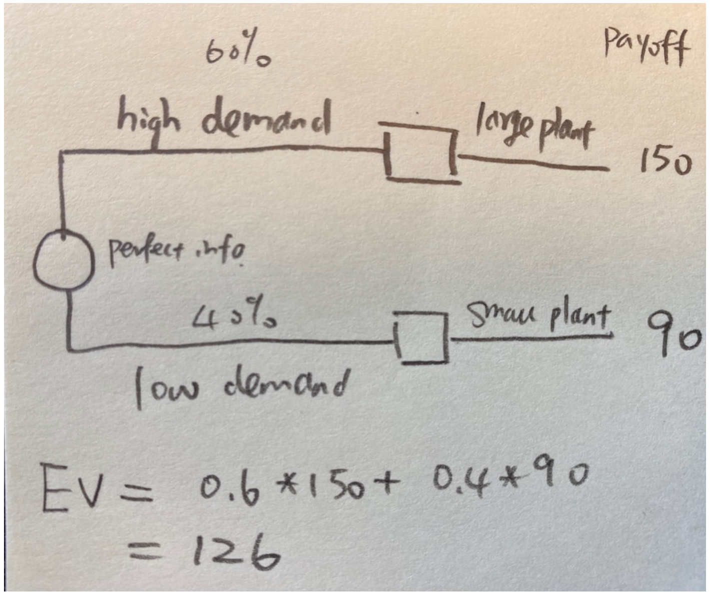
	demand	
Factory size	High	Low
Large	175	95
Small	125	105

Draw decision tree



Part 2: Value of Perfect Information

- Before we decide to build the factory, a consultant company provides **perfect information** about the demand.
- Draw a decision tree corresponding to this scenario. What is the expected value (EV)?



Value of information

Perfect information: 126

With good Survey

With bad Survey

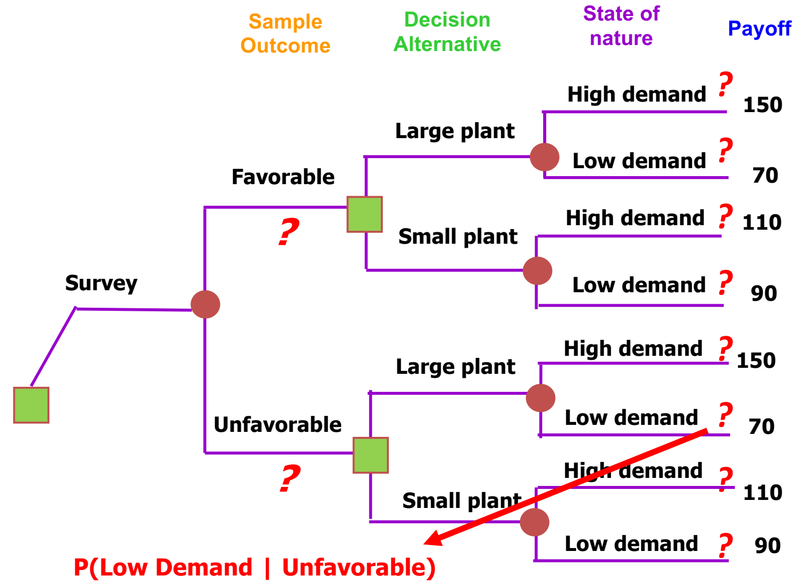
No survey: 118.0

Value of perfect information = 126-118=8

Part 3: Decision with Sample Information

- Now the consultant company does not have the perfect information, but it can conduct a consumer attitude survey. The survey can indicate **favorable** or **unfavorable** attitudes toward the new car.
- Historically, when the demand was high, the survey response was favorable in 80% of the cases, and when the demand was low, the survey response was unfavorable in 75% of the cases.
- Draw the decision tree for this scenario

Decision Tree with Survey



Joint and Marginal Probabilities

	Favorable	Unfavorable	total
High demand			
Low demand			
total			

P(high demand) = 0.6; P(low demand) = 0.4
P(favorable | high demand) = 0.8
P(unfavorable | low demand) = 0.75

- P(high demand | favorable)

= P(high demand and favorable) / P(favorable)

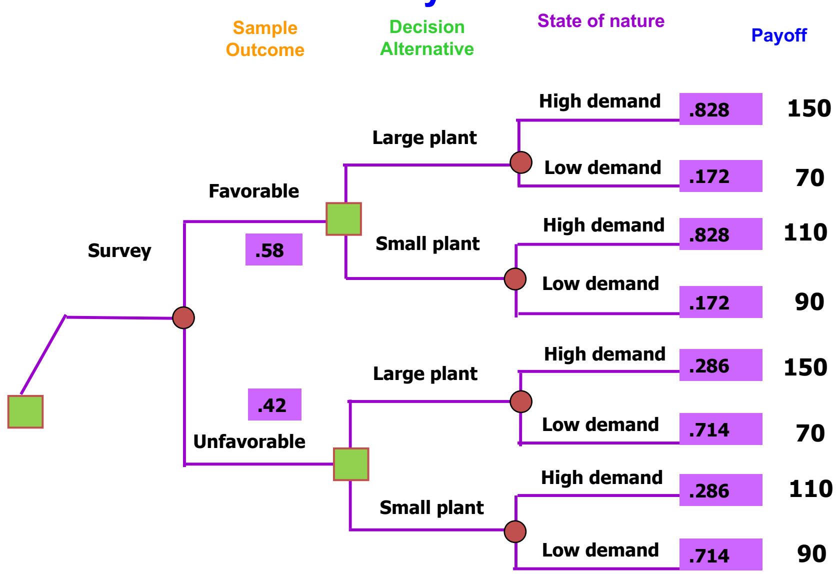
= 0.48 / 0.58 = 0.828

- P(low demand | favorable) =

- P(high demand | unfavorable)=

- P(low demand | unfavorable) =

Decision Tree with Survey



Folding Back Decision Trees

- After the tree is drawn, we fold it back to determine the optimal strategy.
- For decision trees, “forward” is left to right, “backward” is right to left.
- When a decision tree is folded back, it means that the analysis is done right to left.
  - Chance nodes: calculate the expected monetary value of the node using payoffs and associated probabilities for each branch.
  - Decision nodes: select the best option (one with highest EV) and cross out the others.

Summary

- Strategy:
  - If the survey response is favorable, we build a large facility.
  - If the survey response is unfavorable, we build a small facility.
- EV with Survey = ______
- EV w/o Survey = 118.0

Expected Value of Sample Information (EVSI)

Perfect information: 126

With this sample information 119.2

No survey: 118.0

Expected Value of sample information = 119.2-118=1.2